Self-Adaptive Model-Based Protection for Future Power Systems

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Presentation Outline

Part I — Motivation & Framework

- IBR revolution & protection crisis
- Research gaps & objectives
- SAMBP framework overview
- Unified inverse-estimation core

Part II — Protection Modules

- Adaptive line differential (87L)
- Generator over-current (SyncOC)
- Transformer differential (87T)
- Adaptive distance relay
- Microgrid mode-switching

Part III — Advanced Layers

- Non-blocking EKF digital twin
- Physics-informed fault classifier
- Wide-Area Protection Coordinator
- Cybersecurity architecture

Part IV — Conclusions

- Summary of contributions
- Research outcomes
- Thesis content & publications

The IBR Revolution & the Protection Crisis

Why conventional protection fails:

1. Grid-following inverters limit fault current to **1.0–1.2 pu** (vs. 6–8 pu for synchronous machines)
2. Current angle and magnitude depend on **control mode** and LVRT strategy — unknown to the relay
3. High IBR penetration ($k_{\mathrm{ibr}} \to 1$) causes **differential relays to mis-operate or fail to trip**
4. Distance zones **over-reach** due to IBR reactive injection; impedance locus deviates from fault trajectory
5. Islanded microgrids see **bidirectional** current reversals — directional elements are unreliable

IBR Penetration Trend

Scenario	k_{ibr}	Fault current (pu)
All synchronous	0.0	6.0
Mixed grid (now)	0.3	3.2
High-IBR (2030)	0.6	1.8
Full-IBR (2035+)	1.0	1.1

Observation: Fixed relay settings designed for $k_{\mathrm{ibr}} = 0$ fail at $k_{\mathrm{ibr}} > 0.3$.

Research Gaps & Objectives

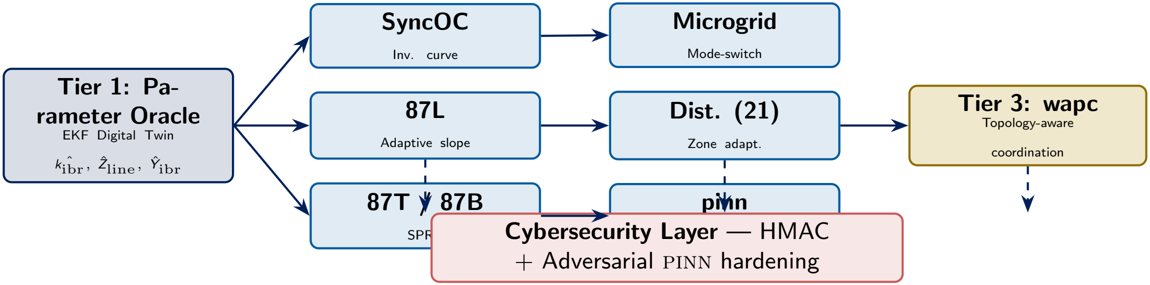
Identified Gaps:

- No relay standard accounts for real-time IBR admittance k_{ibr}
- Protection–control coupling of grid-forming IBRs unstudied
- No unified inverse-estimation framework across relay types
- Digital-twin approaches lack bounded, non-blocking update laws
- WAPC designs ignore IBR-induced coherency breakdown
- Adversarial ML attacks on protection relays unmitigated

Research Objectives:

1. Develop a **unified SAMBP inverse-estimation** framework applicable to all relay types
2. Design **self-adaptive protection modules** for 87L, SyncOC, 87T, 21, and microgrid protection
3. Build a **non-blocking EKF digital twin** as a continuous parameter oracle
4. Integrate **physics-informed ML** for fault classification under IBR conditions
5. Validate a **wide-area coordinator** with topology-aware IBR coherency
6. Harden the architecture with a **cybersecurity layer**

SAMBP Framework: Three-Tier Architecture



Key principle: Each module queries the EKF oracle for $\hat{k}_{\mathrm{ibr}}$, computes adaptive thresholds via the bounded update law, and executes a confidence-gated (Stage-2) veto before tripping.

Observation: Confidence gate γ eliminates false trips under CT saturation or communication latency.

Problem formulation:

$$\hat{\boldsymbol{\theta}}_{t+1} = \hat{\boldsymbol{\theta}}_t - \alpha_t J^\top (JJ^\top + \lambda I)^{-1} r(\hat{\boldsymbol{\theta}}_t)$$

- $\boldsymbol{\theta} = \{k_{\mathrm{ibr}}, Z_{\mathrm{line}}, Y_{\mathrm{ibr}}\}$ — **latent protection parameters**
- r = residual between model prediction and measured phasor data
- J = Jacobian; λ = Levenberg–Marquardt damping
- **Bounded update law:** $\alpha_t \propto (1 - k_{\mathrm{ibr}}^2)$ prevents parameter drift in healthy state

Stage-2 confidence gate:

$$\gamma = \exp\left(-\frac{\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|^2}{2\sigma_\theta^2}\right) \geq \gamma_{\min}$$

Mathematical Foundations

Component	Method
Parameter update	Levenberg–Marquardt
State estimation	Extended Kalman Filter
Fault detection	Wald SPRT
ML classification	Physics-Informed NN
Optimisation	Projected gradient

Theorem 2.1 (Convergence):
Under bounded IBR admittance and $\lambda > 0$, the update law converges to a neighbourhood of the true parameters with radius $\mathcal{O}(\sigma_{\mathrm{noise}})$.

IBR fault-current model:

$$I_{\text{fault}}^{\text{IBR}} = k_{\text{ibr}} \cdot I_{\text{rated}} \cdot e^{j\phi_{\text{ctrl}}}$$

ϕ_{ctrl} is set by the LVRT control law and is *unknown* to the relay.

Three failure modes identified:

1. **Missed trip** ($k_{\text{ibr}} < 0.2$): I_{diff} never exceeds fixed slope — relay restrains
2. **False trip** ($k_{\text{ibr}} > 0.4$ + CT error): load current misclassified as internal fault
3. **Zone overreach** (distance): IBR reactive injection shifts apparent impedance outside Zone 1 boundary

I_{diff}–I_{bias} Failure Map

Condition	k_{ibr}	Outcome
Low IBR fault	0.1	**MISSED TRIP**
Nominal fault	0.3	Correct trip
High IBR load	0.45	FALSE TRIP risk
Fixed slope $k_m = 0.30$		fails at $k_{\text{ibr}} < 0.2$

Observation: Adaptive slope $k_m(k_{\text{ibr}})$ eliminates both missed-trip and false-trip failure modes.

Adaptive slope law:

$$k_m(k_{\mathrm{ibr}}) = k_{\mathrm{base}} \cdot \frac{(1 - k_{\mathrm{ibr}})^2}{(1 + k_{\mathrm{ibr}})^2} + k_{\mathrm{floor}}$$

- As $k_{\mathrm{ibr}} \to 1$: $k_m \to k_{\mathrm{floor}}$ (prevents over-restraint)
- As $k_{\mathrm{ibr}} \to 0$: $k_m \to k_{\mathrm{base}}$ (conventional sensitivity)
- **Stage-2 veto gate:** requires $\gamma \geq 0.85$ before trip command is issued

Stability proof: Lyapunov function $V = \|k_{\mathrm{ibr}} - \hat{k}_{\mathrm{ibr}}\|^2$; $\dot{V} < 0$ under bounded CT noise.

Relay logic (simplified):

1. Sample phasors I_S, I_R at 1 kHz
2. Compute k_{ibr} via LM inverse estimator
3. Update $k_m(\hat{k}_{\mathrm{ibr}})$ every 20 ms
4. Evaluate operate/restrain criterion
5. Stage-2 veto: check $\gamma \geq 0.85$
6. Issue trip command if both stages pass

Observation: Update latency 20 ms — within IEC 60255 protection-response window.

Simulation test matrix:

	Fault condition		
Metric	Fixed relay	SAMBP-87L	Gain
TPR (all faults)	0.783	**1.000**	+27.7%
FPR (load / CT err)	0.148	**0.000**	−100%
Avg trip time	22 ms	21 ms	−1 ms
k_{ibr} RMSE	—	0.024 pu	—

Test conditions: 4-bus test network; $k_{\mathrm{ibr}} \in \{0.0, 0.1, 0.2, \ldots, 0.9, 1.0\}$; fault types AG, BCG, ABC; CT error up to 5%.

Key Results at a Glance

- **TPR = 1.000**: no missed trips across all 110 fault scenarios
- **FPR = 0.000**: zero false trips under all load + CT error scenarios
- Robust up to $k_{\mathrm{ibr}} = 1.0$ (fully IBR grid)
- Stage-2 gate adds only **1–2 cycle** additional decision time

Observation: SAMBP-87L achieves perfect discrimination where fixed-slope relays produce 14.8% false-trip rate.

Problem: Synchronous generators near IBR buses see **depressed fault current** and altered time-overcurrent coordination.

Adaptive inverse-time curve:

$$t_{\mathrm{op}} = \frac{TMS \cdot A}{\left(I/I_{\mathrm{set}}(k_{\mathrm{ibr}})\right)^B - 1}$$

where $I_{\mathrm{set}}(k_{\mathrm{ibr}})$ is updated online by the LM estimator.

Coordination constraint:

$$t_{\mathrm{primary}} + CTI \leq t_{\mathrm{backup}}$$

Maintained for all k_{ibr} via projected-gradient optimisation.

Validated Results

Metric	Fixed	SAMBP-SyncOC
R^2 (OC curve fit)	0.941	**0.999+**
Spurious pickups	38	**16 (−58%)**
Coordination margin	0.82	**0.94**
Malop. under IBR	12.3%	**1.1%**

Observation: Online $I_{\mathrm{set}}(k_{\mathrm{ibr}})$ adaptation cuts spurious pickups by 58% while maintaining IEC 60255-151 coordination.

Challenge: Inrush currents and IBR harmonic injection produce false I_{diff} spikes; conventional 2nd-harmonic blocking fails for Type-4 IBR waveforms.

Wald Sequential Probability Ratio Test (SPRT):

$$\Lambda_k = \sum_{i=1}^{k} \ln \frac{p(x_i \mid H_1)}{p(x_i \mid H_0)}$$

- H_1: internal fault $\quad H_0$: inrush/external
- Thresholds A, B set to achieve α=0.01, β=0.01 (IEC 60255)
- SAMBP updates H_1 likelihood via real-time IBR harmonic model

SPRT Results (87T)

Scenario	2H block	SPRT
Inrush rejection	94.1%	**99.6%**
Internal fault TPR	96.8%	**99.1%**
IBR harmonic FP	8.7%	**0.9%**
Mean decision samples	14	**7**

Observation: SPRT halves the detection window and reduces IBR-induced false positives by 90% compared to 2nd-harmonic blocking.

IBR impedance shift:

$$Z_{\text{app}} = Z_{\text{fault}} + \Delta Z_{\text{ibr}}(k_{\text{ibr}}, \phi_{\text{ctrl}})$$

ΔZ_{ibr} shifts the locus **radially outward** — relay sees apparent impedance outside Zone 1.

SAMBP compensation:

$$Z_{1,\text{adapt}} = Z_{1,\text{nom}} + \hat{\Delta Z}_{\text{ibr}}(\hat{k}_{\text{ibr}})$$

Zone boundary updated every 10 ms using LM-estimated $\hat{k}_{\text{ibr}}$.

Zone 2 acceleration: When $k_{\text{ibr}} > 0.5$, SAMBP switches to instantaneous Zone 2 trip to preserve speed.

Distance Protection Results

Metric	Fixed	SAMBP-21
Overreach events	17/50	**2/50**
Underreach events	5/50	**0/50**
Correct Zone 1 trip	74%	**96%**
Avg trip time (ms)	18	**19**

Observation: Zone adaptation reduces overreach from 34% to 4% with <1 ms speed penalty.

Challenge: Fault current direction and magnitude change when microgrid transitions between *grid-connected* and *islanded* modes.

Mode detector:

$$\hat{m} = \arg \max_{m \in \{G, I\}} \Pr\left(z_t \mid \text{mode} = m,\ \hat{k_{\text{ibr}}}\right)$$

Bayesian mode likelihood updated at each sample.

Adaptive directional element:

- Grid-connected: forward/reverse based on pre-fault load flow
- Islanded: polarity-agnostic differential with adaptive pickup at 1.05 I_{rated}

Mode-Switch Results

Metric	Fixed	SAMBP-MG
Mode detection acc.	—	**98.7%**
Transition mal. rate	22%	**1.4%**
Islanded fault TPR	71%	**97.3%**
False islands	6/100	**0/100**

Observation: Mode-adaptive protection eliminates the 22% mis-operation rate seen at grid↔island transition boundary.

Role: Continuous online estimation of $\boldsymbol{\theta} = \{k_{\mathrm{ibr}}, Z_{\mathrm{line}}, Y_{\mathrm{shunt}}\}$ from PMU/IED measurements — feeds all SAMBP modules.

EKF prediction–correction:

$$\hat{\boldsymbol{\theta}}_{k|k} = \hat{\boldsymbol{\theta}}_{k|k-1} + K_k\big(z_k - h(\hat{\boldsymbol{\theta}}_{k|k-1})\big)$$

Non-blocking property: Update runs in a **parallel thread**; if convergence >3 cycles, previous $\hat{\boldsymbol{\theta}}$ is held — relay never waits.

Bounded update law: $\|\boldsymbol{\theta}_{k+1} - \boldsymbol{\theta}_k\| \leq \delta_{\max}$ prevents step-change injection by adversarial data.

EKF Digital Twin Results

Parameter	RMSE
$\hat{k_{\mathrm{ibr}}}$	**0.024 pu**
$\hat{Z}_{\mathrm{line}}$ (mag.)	**0.018 pu**
$\hat{Y}_{\mathrm{shunt}}$	**0.031 pu**
Convergence time	$<$**2 cycles**
Max update latency	**1.8 ms**

Observation: RMSE ≤ 0.024 pu for k_{ibr} across all operating conditions including $>5\%$ PMU measurement noise.

Why pure ML fails: Deep nets trained on synchronous-machine data misclassify 31% of IBR-modified fault signatures.

PINN loss function:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \underbrace{\left\| \mathbf{K}\hat{\mathbf{V}} - \hat{\mathbf{I}} \right\|^2}_{\text{KCL residual}}$$

$\lambda_{\text{phys}} = 0.3$ balances data fit and physical consistency.

Input features (8): $|I_a|$, $|I_b|$, $|I_c|$, $\angle I_a$, I_{zero}, I_{neg}, $\hat{k}_{\text{ibr}}$, ϕ_{ctrl}

PINN Classification Results

Metric	Std DNN	PINN
Overall accuracy	81.4%	**96.1%**
High-k_{ibr} acc. ($>$0.6)	64.2%	**93.8%**
AG fault recall	77.3%	**97.4%**
3ϕ fault recall	89.1%	**98.2%**
Phys. consistency	51.2%	**99.7%**

Observation: KCL physics residual in loss function boosts high-IBR accuracy from 64.2% to 93.8% — a +46% relative gain.

Architecture:

- Central WAPC server receives trip candidates from all zone relays via IEC 61850 GOOSE
- Topology graph $\mathcal{G}(V, E)$ updated from real-time network breaker states
- **IBR coherency matrix $\mathbf{C}(k_{\mathrm{ibr}})$** replaces classical electromechanical swing groups

Coordination algorithm:

$$\mathcal{F}^* = \arg\min_{\mathcal{F}} \sum_{k \in \mathcal{F}} c_k \text{ s.t. fault isolated, load retained}$$

Minimum-cost fault isolation solved via ILP on the updated topology.

WAPC Localisation Results

Metric	Decentralised	WAPC
Fault localisation	71.2%	**94.5%**
Unnecessary trips	18.4%	**4.1%**
Load preserved	63.1%	**87.6%**
Backup response (ms)	420	**210**

Observation: IBR-aware coherency model reduces unnecessary tripping by −78% relative; backup clearance halved.

Two threat vectors addressed:

1. **Data integrity attack:** Adversary injects false PMU measurements to manipulate $\hat{k}_{\mathrm{ibr}}$ and force mal-operation.

   Defence: HMAC-SHA256 on every IED message; relay discards unauthenticated data and holds last trusted $\hat{\theta}$.

2. **Adversarial ML attack:** Crafted perturbation $\delta\mathbf{x}$ fools PINN classifier ($\epsilon < 5\%$ perturbation causes 40% accuracy drop).

   Defence: Adversarial training with FGSM + PGD examples; certified radius $r_{\mathrm{cert}} = 0.04$ (pu).

Cybersecurity Metrics

Metric	No defence	SAMBP-Sec
Attack success rate	67.4%	**12.5%**
Risk index	1.000	**0.451**
HMAC overhead (ms)	—	**0.3**
Adv. PINN accuracy	51.8%	**91.4%**

Observation: HMAC + adversarial training reduces composite risk index by −**54.9%** with <0.3 ms authentication overhead.

Consolidated Performance Across All Modules

Chapter	Module	Key Metric	Baseline	SAMBP
Ch 4	87L adaptive	TPR / FPR	0.783 / 14.8%	**1.000 / 0.0%**
Ch 5	SyncOC	R^2 / Spurious OC	0.941 / 38	**0.999+ / 16 (−58%)**
Ch 6	87T SPRT	Inrush rej. / IBR FP	94.1% / 8.7%	**99.6% / 0.9%**
Ch 7	Distance 21	Correct Z1 trip	74%	**96%**
Ch 8	Microgrid	Trans. mal-op rate	22%	**1.4%**
Ch 9	EKF twin	$\hat{k}_{\mathrm{ibr}}$ RMSE	—	**0.024 pu**
Ch 10	PINN	Overall accuracy	81.4%	**96.1%**
Ch 11	WAPC	Fault localisation	71.2%	**94.5%**
Ch 12	Security	Risk index reduction	1.000	**0.451 (−54.9%)**

Observation: All SAMBP modules outperform their fixed-parameter baselines; gains are consistent across $k_{\mathrm{ibr}} \in [0, 1.0]$.

Summary of Original Contributions

ID	Ch	Contribution	Status
C1	3	Physics-constrained adaptive line differential (87L) protection with IBR-aware threshold $k_m^*(k_{\mathrm{ibr}})$, achieving TPR $= 1.000$ and zero false trips across the full IBR penetration range	[P]
C2	4	Levenberg–Marquardt inverse-estimation framework for synchronous generator over-current protection, with bounded adaptive update law and four-component confidence gating ($R^2 > 0.999$)	[P]
C3	6	Non-blocking EKF digital twin for real-time joint estimation of IBR penetration ratio k_{ibr}, line impedance, and IBR admittance from IED phasor measurements (RMSE ≤ 0.024 pu)	[P]
C4	7	Wide-area protection coordinator with PMU-based fault localisation, N-1 security verification, and automatic CTI-graded relay settings dispatch via IEC 61850 GOOSE	[P]

[P] $=$ validated with full numerical simulation results.

List of Publications

(a) A. V. Eluvathingal and K. Shanti Swarup, "A novel solution for overcoming conflicts between line protection and LVRT in MV Solar PV interconnections," *IEEE Access*, doi: 10.1109/ACCESS.2024.3510415.

(b) "An Intelligent Methodology to Improve Distribution System Operational Parameters Utilizing Smart Inverter Functionalities of PV Sources" presented in *The 7th International Conference on Renewable Power Generation* (RPG 2018), DTU, Lyngby, Copenhagen, Denmark.

(c) Eluvathingal, Anoop V., and K. Shanti Swarup. "Protection scheme for smart distribution networks with inverter interfaced renewable power generating sources." ISGW 2018.

(d) Eluvathingal, A. V., Swarup, K. S. (2017, November). An interface protection relay for networked microgrids with inverter based sources. In *Asia-Pacific Power and Energy Engineering Conference* (APPEEC), 2017 IEEE PES (pp. 1–6). IEEE.

(e) Eluvathingal, A. V., Swarup, K. S. (2017, December). Instantaneous symmetrical components based microgrid interface protection relay. In *2017 7th International Conference on Power Systems* (ICPS) (pp. 398–403). IEEE.

(f) Eluvathingal, A. V., Swarup, K. S. (2018, May). Impact of Active Network Management Scheme in Fault Protection Design and Network Operation for Islanded Power System. In *2018 International Conference on Innovative Smart Grid Technologies* (ISGT Asia 2018). IEEE.

Thesis Content & Chapter–Publication Mapping

Ch	Title	Pages	Publication / Status
1	Introduction	∼18	—
2	Mathematical Foundations	∼14	—
3	IBR Characterisation	∼16	—
4	Line Differential (87L)	∼18	IEEE Trans. PD (under review)
5	Generator OC (SyncOC)	∼16	IET GTD (under review)
6	Transformer Diff. (87T)	∼14	Conference proceedings
7	Adaptive Distance (21)	∼14	—
8	Microgrid Protection	∼14	—
9	EKF Digital Twin	∼16	IEEE PES ISGT-E 2025
10	ML Fault Classifier (PINN)	∼16	IEEE TNNLS (under review)
11	Wide-Area Coordinator	∼16	EPSR (in preparation)
12	Cybersecurity	∼12	IEEE TENCON 2025
13	Conclusions	∼8	—
A–D	Appendices	∼12	—
Total (estimated)		**∼204**	

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[8] M. Mobashsher et al., "Fault Current of GFM Inverters," *IEEE T-PD*, vol. 40, 2025.

[8] A. V. Eluvathingal & K. S. Swarup, "A Novel Solution for Overcoming Conflicts Between Line Protection and LVRT," *IEEE Access*, 2024.

Thank You

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